

Lab 4 plus

NAME: GUI KAH SIN
MATRIC NO: A23CS0080
SECTION: 12

*Lab 4 plus is a combination of Lab 4 and an extra activity on ARP.

Packet Tracer Simulation – Exploration of ARP and Switch Table Communications

Objectives

- To explore ARP and switching operations.

Introduction

The topology is given to you. All IP addresses have been assigned to all devices. Please follow each step in sequence.

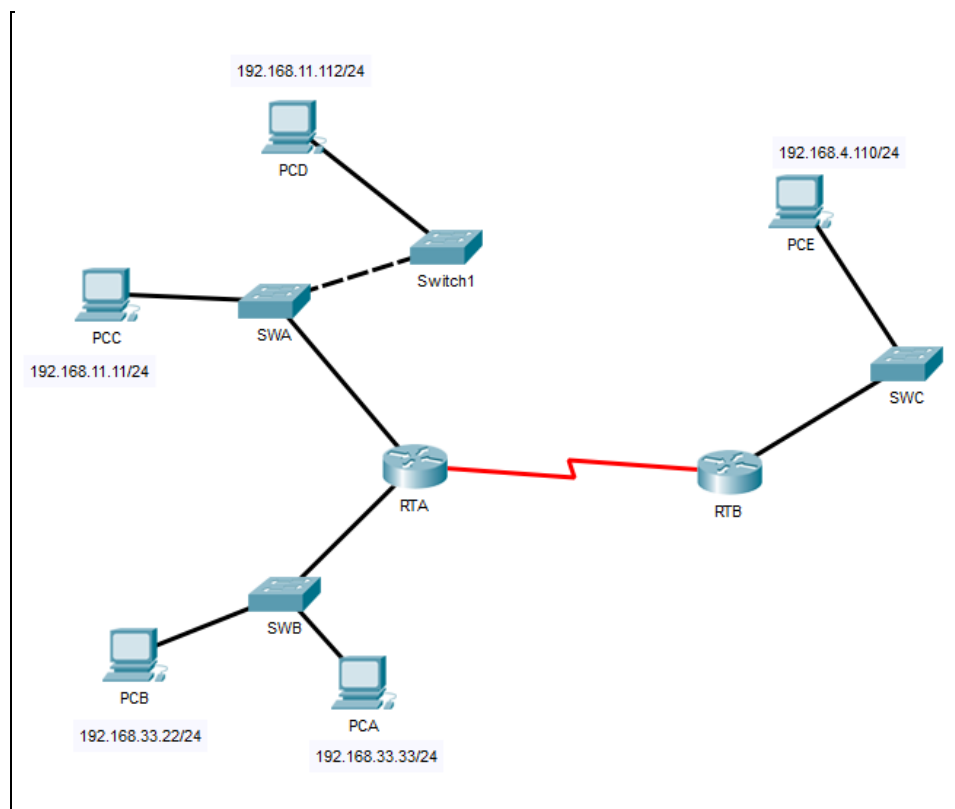


Figure 1

Part 1: Review the topology

Step 1: Perform the following tasks.

- a. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.11.1      -          0002.4A00.0E91  ARPA   FastEthernet1/0
Internet  192.168.33.1      -          000C.CF0C.593A  ARPA   FastEthernet0/0
RTA#
```

- b. At Router RTB, enter the CLI. At the command prompt type the commands as in Figure 2. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
Protocol  Address          Age (min)  Hardware Addr  Type   Interface
Internet  192.168.4.1      -          0001.977A.B614  ARPA   FastEthernet0/0
RTB#
```

- c. At Switches SWA, SWAB and SWC, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp
SWA#show mac-address-table
```

SWA:

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type        Ports
----    -
1       0002.4a00.0e91    DYNAMIC     Fa0/1
1       000c.8546.7d85    DYNAMIC     Fa1/1
SWA#
```

SWB:

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
SWB#
```

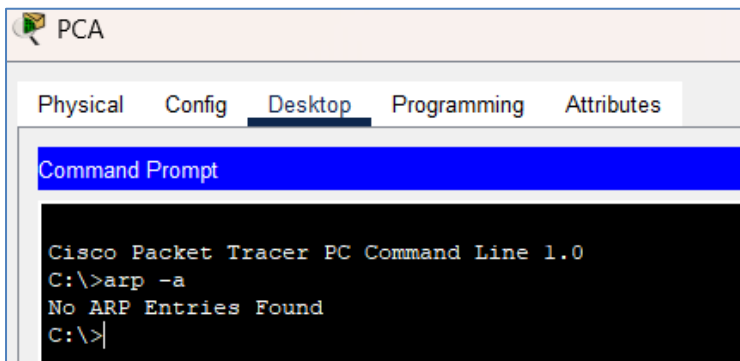
SWC:

```
SWC>enable
SWC#show arp

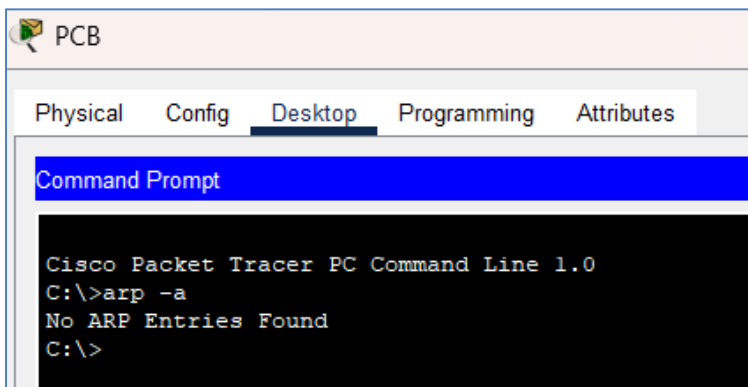
SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
SWC#
```

- d. At PCA, click on the PC icon, and then choose Desktop-Command Prompt. At the command prompt type **arp -a** and click enter. Snap the results after the last command and paste it here. Do this to all PCs in the topology.

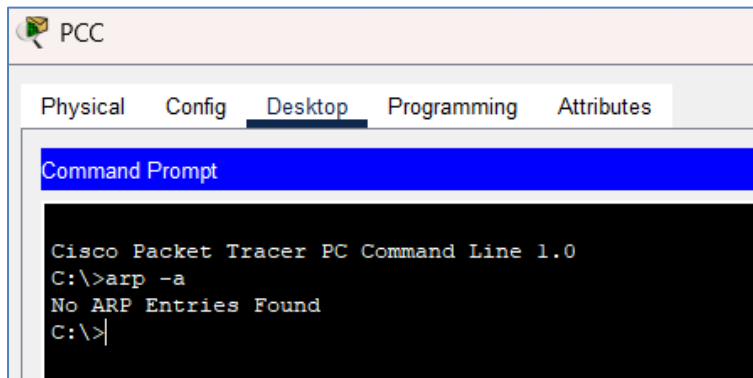
PCA:



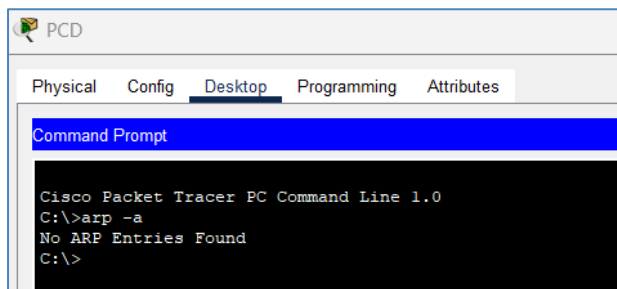
PCB:



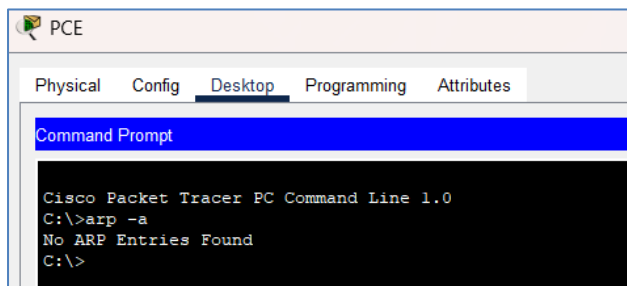
PCC:



PCD:



PCE:



- e. What are your thoughts on the results?

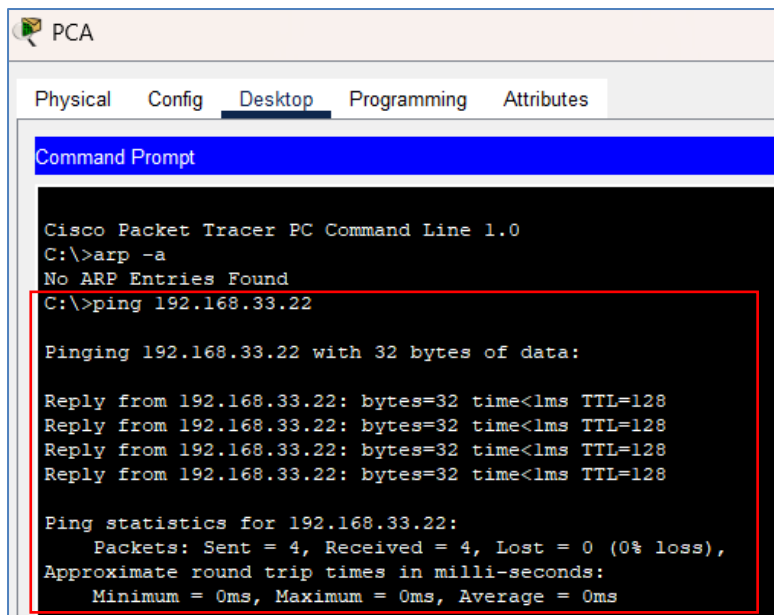
The results show no ARP entries, indicating the PC has not communicated with other devices, there are no ARP requests or responses.

Part 2: Generate Network Traffic

Step 1: Generate traffic between PCA and PCB.

In the command prompt Perform the following tasks task to reduce the amount of network traffic viewed in the simulation.

- Click **PCA** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command. This may take a few seconds.



PCA

Physical Config Desktop Programming Attributes

Command Prompt

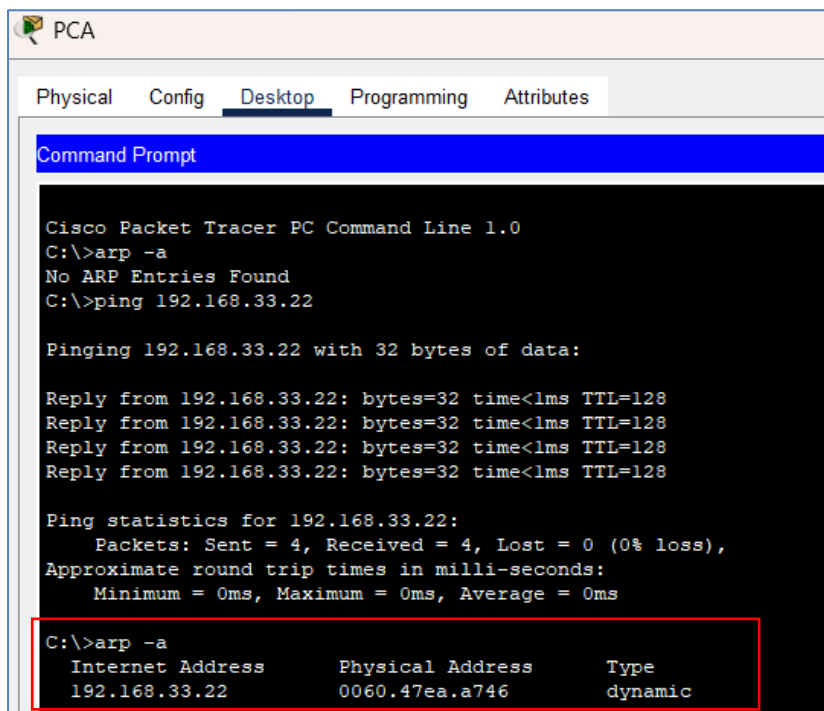
```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

- c. In the Command prompt of PCA, type **arp -a**. Paste the result of this command here.



PCA

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>ping 192.168.33.22

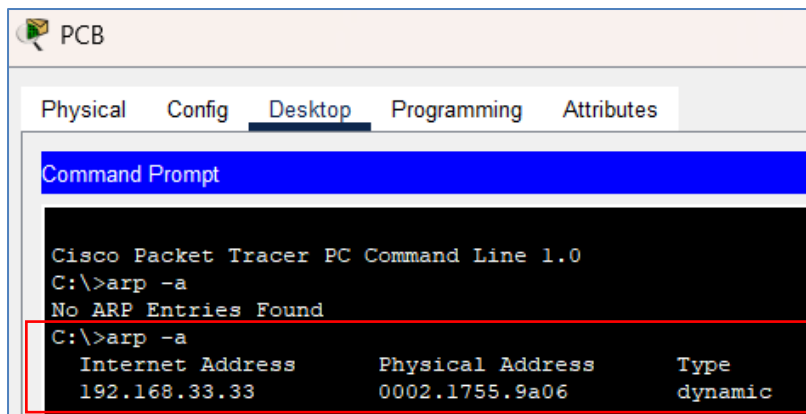
Pinging 192.168.33.22 with 32 bytes of data:

Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128
Reply from 192.168.33.22: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.33.22:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

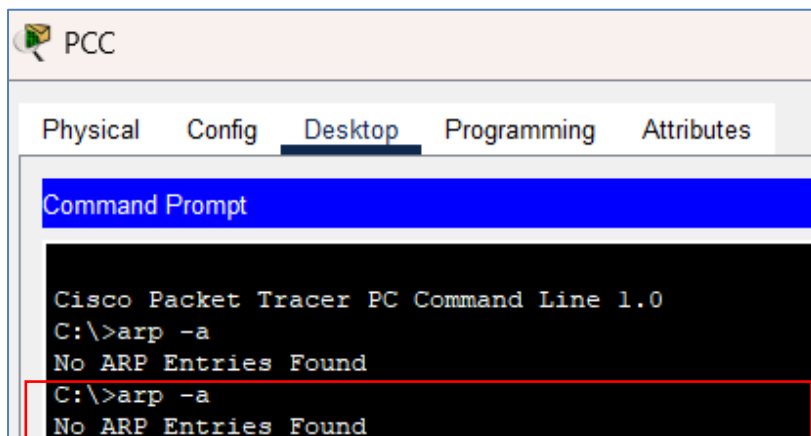
C:\>arp -a
Internet Address      Physical Address      Type
192.168.33.22         0060.47ea.a746        dynamic
```

- d. In the Command prompt of PCB, type **arp -a**. Paste the result of this command here

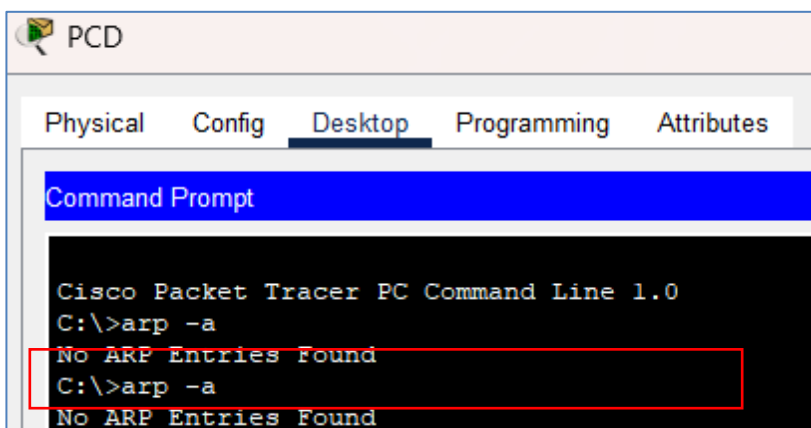


- e. In the Command prompt of PCC, PCD and PCE, type **arp -a**. Paste the result of this command here.

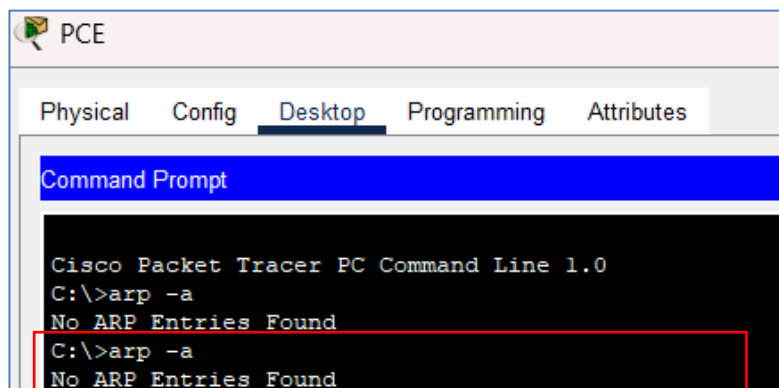
PCC:



PCD:

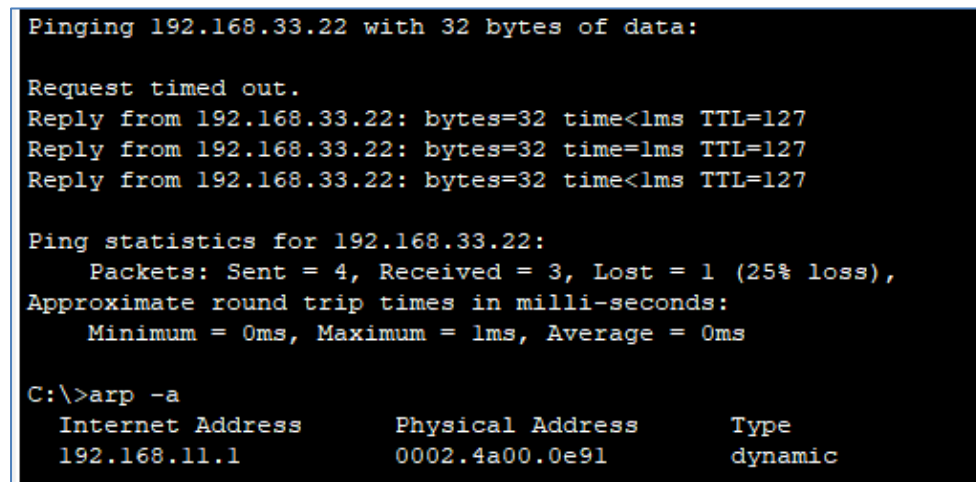


PCE:



Step 2: Generate traffic between PCC to all other PC except PCA.

- Click **PCC** and click the Desktop tab > Command Prompt.
- Enter the **ping 192.168.33.22** command (ping to PCB). Then type **arp -a**. Paste the result after these commands here.



- c. Enter the **ping 192.168.11.112** command (ping to PCD). Then type **arp -a**. Paste the result after these commands here.

```
Pinging 192.168.11.112 with 32 bytes of data:

Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time=1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128
Reply from 192.168.11.112: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.11.112:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

- d. Enter the **ping 192.168.4.110** command (ping to PCE). Then type **arp -a**. Paste the result after these commands here.

```
C:\>ping 192.168.4.110

Pinging 192.168.4.110 with 32 bytes of data:

Reply from 192.168.4.110: bytes=32 time=1ms TTL=126
Reply from 192.168.4.110: bytes=32 time=16ms TTL=126
Reply from 192.168.4.110: bytes=32 time=12ms TTL=126
Reply from 192.168.4.110: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.4.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 16ms, Average = 7ms

C:\>arp -a

Internet Address      Physical Address      Type
192.168.11.1          0002.4a00.0e91       dynamic
192.168.11.112        0001.6462.0278       dynamic
```

- e. Discuss the results you got from all the commands on PCC.

The results show no initial ARP entries on PCC as it had not pinged or communicated with other devices. After PCC pinging PCB, PCD and PCE, ARP requests resolved their MAC addresses, populating table with entries. This shows ARP's role in dynamically mapping IP to MAC addresses during communication.

- f. At Router RTA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTA>enable
RTA#show arp
```

```
RTA>enable
RTA#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.11.1 - 0002.4A00.0E91 ARPA FastEthernet1/0
Internet 192.168.11.11 29 00D0.D39A.C0D9 ARPA FastEthernet1/0
Internet 192.168.33.1 - 000C.CF0C.593A ARPA FastEthernet0/0
Internet 192.168.33.22 29 0060.47EA.A746 ARPA FastEthernet0/0
RTA#
```

- g. At Router RTB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
RTB>enable
RTB#show arp
```

```
RTB>enable
RTB#show arp
Protocol Address Age (min) Hardware Addr Type Interface
Internet 192.168.4.1 - 0001.977A.B614 ARPA FastEthernet0/0
Internet 192.168.4.110 26 0060.702D.7C08 ARPA FastEthernet0/0
RTB#
```

Step 3: Switch MAC address table.

- a. At Switch SWA, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
```

```
SWA>enable
SWA#show arp

SWA#show mac-address-table
Mac Address Table
-----
Vlan Mac Address Type Ports
---
1 0002.4a00.0e91 DYNAMIC Fa0/1
1 000c.8546.7d85 DYNAMIC Fa1/1
SWA#
```

- b. At Switch SWB, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
```

```
SWB>enable
SWB#show arp

SWB#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       000c.cf0c.593a   DYNAMIC Fa0/1
SWB#
```

- c. At Switches SWC and Switch1, enter the CLI. At the command prompt type the following commands. Snap the results after the last command and paste it here.

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
```

```
SWC>enable
SWC#show arp

SWC#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.977a.b614   DYNAMIC Fa0/1
SWC#
```

```
Switch>enable
Switch#show arp

Switch#show mac-address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0005.5e95.187b   DYNAMIC Fa0/1
Switch#
```

- d. Do switches use arp table? (Y/N)

N

- e. Explain your answer in (d) **Hint: the answer may surprise you. Google for the explanation. It is not part of NetComm syllabi, it is just for knowledge..*

Switches work with MAC addresses, not IP addresses. They use a MAC address table to forward data. ARP tables, which map IPs to MAC addresses, are used by devices like routers, not Layer 2 switches. Only Layer 3 switches use ARP tables for routing. When I typed “show ip routes” on the switch, it showed error message, which indicates it is a Layer 2 switches.

- f. What information does the command **show mac-address-table** gives?

The “show mac-address-table” command displays the MAC addresses on the switch along with their associated ports, and VLANs. It also indicates whether the MAC addresses were learned dynamically or configured manually. This table helps the switch forward data to the correct ports based on MAC addresses.

Part 3: Attach wireless lab results.

In this part, you will use Lab 4 .pka file.

Step1: Change the filename of Lab 4.

- Change the Lab 4 filename to include your name. **Example: Lab4AliAhmad.pkt*
- Go through the instructions. As you complete the tasks, you will see the bottom right hand corner of the pkt file increase in completion percentage, until you get 100/100.

The screenshot displays the Packet Tracer interface. On the left, a network topology is shown with a central switch (S1) connected to a router (WRS2) and three PCs (PC1, PC2, PC3). A subinterface table is visible:

Subinterfaces	
G0/0.10	172.17.10.1/24
G0/0.20	172.17.20.1/24
G0/0.88	172.17.88.1/24

On the right, the 'Packet Tracer – Configuring Wireless LAN Access' activity wizard is open, showing an 'Addressing Table':

Device	Interface	IP Address	Subnet Mask	Default Gate
R1	G0/0.10	172.17.10.1	255.255.255.0	N/A
	G0/0.20	172.17.20.1	255.255.255.0	N/A
	G0/0.88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1

At the bottom right of the activity wizard, the completion status is shown as 'Completion: 100/100' with a red box highlighting the progress bar and navigation buttons.

- Once you have completed fully, capture the screen (which includes the filename, the topology and the activity wizard showing completion) and paste it here.

Cisco Packet Tracer - C:\Users\Stein\Downloads\Lab 4 GUI KAH S1N.pka - Guest - 2025-01-21 23:16:47

File Edit Options View Tools Extensions Window Help

Logical Physical x 488, y 6 Root 00:15:00

Subinterfaces

- G0/0/10 172.17.10.1/24
- G0/0/20 172.17.20.1/24
- G0/0/88 172.17.88.1/24

PC1 172.17.10.21/24 VLAN 10

PC2 172.17.20.22/24 VLAN 20

WRS2 172.17.88.25/24 VLAN 88

PC3 172.17.40.100/24

PT Activity: 00:00:30

Packet Tracer – Configuring Wireless LAN Access

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/0/10	172.17.10.1	255.255.255.0	N/A
	G0/0/20	172.17.20.1	255.255.255.0	N/A
	G0/0/88	172.17.88.1	255.255.255.0	N/A
PC1	NIC	172.17.10.21	255.255.255.0	172.17.10.1
PC2	NIC	172.17.20.22	255.255.255.0	172.17.20.1
PC3	NIC	DHCP Assigned	DHCP Assigned	DHCP Assigned
WRS2	NIC	172.17.88.25	255.255.255.0	172.17.88.1

Objectives

- Part 1: Configure a Wireless Router
- Part 2: Configure a Wireless Client
- Part 3: Verify Connectivity

Time Elapsed: 00:00:30 Completion: 100/100

☐ Top ☐ Dock

Time: 00:00:25

Realtime Simulation

Scenario 0

Fire Last Status Source Destination Type Color Time(sec) Periodic Num Edit Delete

PC3 PC2 ICMP 0.000 N 0 (edit) (delete)

PC3 PC1 ICMP 0.000 N 1 (edit) (delete)

Toggle PDU List Window

Copper Straight-Through

25°

ENG US 11:23 PM 21/1/2025